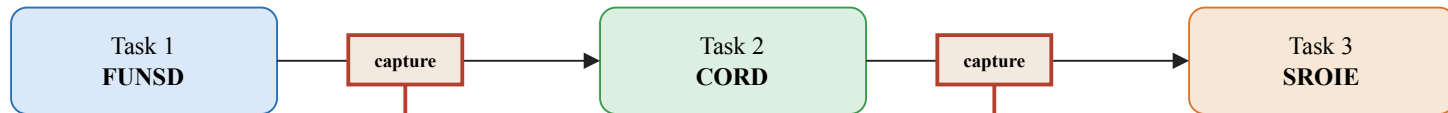


Per-Component Diagnostic Pipeline (Algorithm 3.1)

Modality mask $c \in \{ \text{C1: text-only} \mid \text{C2: image+layout} \mid \text{C3: text+layout} \mid \text{C4: full tri-modal} \}$



naive sequential fine-tuning · classifier head expanded per task

Per-layer linear CKA

$\text{CKA}(\theta^{(t-1)}, \theta^{(t)})$ per layer
measures representational similarity shift

Per-group Fisher Information

$F_g = \text{diag. Fisher per param group}$
groups: text / layout / visual / fusion / ffn / head

Forgetting matrix $R[i, j]$

entity-F1 of task j evaluated after task i
AA / BWT / AF derived from $R[i, j]$

Hypothesis: H_1 — forgetting concentrates in specific components; H_0 (null) — uniform across all. Decision at Week 4 selects CL method.

→ sequential task progression

→ capture / fan-out to metrics

□ task-boundary capture marker